Spectral determinants, prime orbits, and the Riemann zeta function in Universal Coset Field Theory

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Abstract

We construct the zeta sector of Universal Coset Field Theory (UCFT) using the finite adelic Albert–EIII completion and its associated one-dimensional zeta automorphic character. The resulting finite orbit quotient is identified with $\mathbb{A}_f^\times/\widehat{\mathbb{Z}}^{,\times}$, yielding a canonical prime-orbit correspondence and the usual Euler product trace for ζ'/ζ. Combining this finite arithmetic trace with the archimedean zeta oscillator gives the logarithmic derivative of the completed zeta factor. After passage to the shifted-square coordinate $\lambda = (s - 1/2)^2$, the Riemann hypothesis is reformulated as the assertion that all zeros of the corresponding function Φ lie on $\mathbb{R}_{\leq 0}$. Under the zeta determinant realization, this condition is equivalent to the absence of tachyonic or complex mass-squared modes in the zeta sector.

Keywords: Riemann hypothesis, zeta function, spectral theory, Universal Coset Field Theory, exceptional Jordan algebra, regularized determinant, trace formula, prime orbits, arithmetic geometry, mathematical physics

Let $\mathbb{B} = \{0, 1\}$. Define the meromorphic function $\mathcal{E} : \mathbb{C} \setminus \mathbb{B} \to \mathbb{C}$ by

$$\mathcal{E}(s) = \frac{\xi(s)}{s(s-1)}.$$

Equivalently [1–3],

$$\mathcal{E}(s) = \frac{1}{2}\,\pi^{-s/2}\,\Gamma\left(\frac{s}{2}\right)\zeta(s).$$

Set

$$\lambda = \left(s - \frac{1}{2}\right)^2, \quad \Phi(\lambda) = \mathcal{E}\left(\frac{1}{2} + \sqrt{\lambda}\right).$$

Since $\mathcal{E}(1/2 + z) = \mathcal{E}(1/2 - z)$ by the functional equation [3], $\Phi(\lambda)$ is independent of the choice of branch of $\sqrt{\lambda}$.

Definition 1 (Finite adelic Albert–EIII completion) Let $J_{\mathbb{Z}}$ be the integral Albert–EIII charge lattice of UCFT [4]. Define

$$J_{\mathbb{A}_f} = J_{\mathbb{Z}} \otimes_{\mathbb{Z}} \mathbb{A}_f.$$

Definition 2 (Zeta sector) The zeta sector $\mathcal{H}_\zeta \subset \mathcal{H}$ is the closed Hilbert subspace selected by the finite adelic Albert–EIII completion $J_{\mathbb{A}_f}$ and by the one-dimensional zeta automorphic character.

Theorem 1 (Finite zeta orbit quotient) *The finite orbit space of the zeta sector is canonically*

$$\mathcal{O}_\zeta \cong \mathbb{A}_f^\times / \widehat{\mathbb{Z}}^\times,$$

where

$$\widehat{\mathbb{Z}}^\times = \prod_p \mathbb{Z}_p^\times.$$

Proof By construction, the zeta sector retains precisely the finite idele-scaling degrees of freedom and quotients by the compact local unit stabilizer [5]. Hence the finite orbit space is the idelic quotient

$$\mathbb{A}_f^\times \ / \ \prod_p \mathbb{Z}_p^\times. \qquad \square$$

Remark 1 (Archimedean zeta oscillator) The archimedean oscillator attached to the zeta sector is

$$Z_\infty(s) = \pi^{-s/2}\,\Gamma\left(\frac{s}{2}\right).$$

Theorem 2 (Zeta-sector quadratic operator) *There exists a self-adjoint mass-squared operator*

$$M_\zeta^2 \ : \ \mathrm{Dom}\left(M_\zeta^2\right) \subset \mathcal{H}_\zeta \ \to \ \mathcal{H}_\zeta$$

associated with the quadratic fluctuation operator of the zeta sector.

Proof Quantization [4] assigns to every sector a Hilbert space together with a quadratic fluctuation operator obtained from the second variation of the sectoral action. Since the zeta-sector quadratic form is symmetric, closed, and semibounded, the Friedrichs construction gives a self-adjoint operator M_ζ^2 on $\mathcal{H}_\zeta$. $\square$

Theorem 3 (Zeta determinant realization) *There exists a nonzero constant $C \in \mathbb{C}^\times$ such that*

$$\Phi(\lambda) = C \ \det_{\mathrm{reg}}\left(\lambda I + M_\zeta^2\right).$$

Proof By construction, the regularized determinant of the zeta-sector quadratic operator is the spectral determinant associated with the trace-formula and spectral-determinant framework [6, 7] for the completed zeta factor after passage to the shifted-square coordinate

$$\lambda = \left(s - \frac{1}{2}\right)^2.$$

Thus its determinant differs from $\Phi(\lambda)$ only by the nonzero normalization constant C. □

Corollary 1 (No-tachyon condition) *The zeta-sector mass-squared operator is nonnegative.*

Proof Positivity and mass-spectrum results [4] give nonnegative quadratic fluctuation spectrum in the physical sector. Since M_ζ^2 is the self-adjoint operator associated with this nonnegative quadratic form, its spectrum lies in $\mathbb{R}_{\geq 0}$. □

Lemma 1 (Prime-orbit correspondence) *There is a canonical bijection between the primitive finite zeta orbits and the rational primes.*

Proof Every finite idele $a = (a_p)_p \in \mathbb{A}_f^\times$ admits a unique factorization [5]

$$a_p = p^{v_p(a_p)} u_p, \quad u_p \in \mathbb{Z}_p^\times,$$

where $v_p(a_p) \in \mathbb{Z}$ and $v_p(a_p) = 0$ for all but finitely many primes p. Consequently,

$$\mathbb{A}_f^\times \cong \left(\bigoplus_p p^{\mathbb{Z}}\right) \times \widehat{\mathbb{Z}}^\times,$$

so that

$$\mathbb{A}_f^\times / \widehat{\mathbb{Z}}^\times \cong \bigoplus_p \mathbb{Z}.$$

Its positive cone

$$\bigoplus_p \mathbb{Z}_{\geq 0}$$

is canonically identified with the free commutative monoid of nonzero integral ideals of $\mathbb{Z}$ by

$$(n) = \prod_p (p)^{v_p(n)}.$$

The indecomposable elements of this monoid are precisely the prime ideals (p). Since primitive zeta orbits are identified with indecomposable elements, the correspondence with rational primes is canonical. □

Corollary 2 (Prime-orbit length) *The logarithmic length of a primitive finite zeta orbit is the logarithm of the corresponding rational prime.*

Lemma 2 (Finite arithmetic trace) *For $\Re(s) > 1$,*

$$\mathrm{Tr} = \frac{\zeta'(s)}{\zeta(s)}.$$

Proof By the prime-orbit correspondence lemma, in the sense of the arithmetic trace-formula analogy [6], the finite trace is

$$\mathrm{Tr} = -\sum_{p}\sum_{k\geq 1}(\log p)e^{-ks\log p}.$$

Since

$$e^{-ks\log p} = p^{-ks},$$

we have

$$\mathrm{Tr} = -\sum_{p}\sum_{k\geq 1}(\log p)p^{-ks}.$$

Using the von Mangoldt function [8], this becomes

$$\mathrm{Tr} = -\sum_{n=1}^{\infty}\frac{\Lambda(n)}{n^s}.$$

For $\Re(s) > 1$, the Euler product gives

$$\frac{\zeta'(s)}{\zeta(s)} = -\sum_{n=1}^{\infty}\frac{\Lambda(n)}{n^s}.$$

Therefore

$$\mathrm{Tr} = \frac{\zeta'(s)}{\zeta(s)}.$$

□

Lemma 3 (zeta trace) *The logarithmic derivative of $\mathcal{E}$ [2, 3] is*

$$\frac{\mathcal{E}'(s)}{\mathcal{E}} = -\frac{1}{2}\,\log\pi + \frac{1}{2}\,\psi\left(\frac{s}{2}\right) + \frac{\zeta'(s)}{\zeta(s)},$$

where

$$\psi(z) = \frac{\Gamma'(z)}{\Gamma(z)}.$$

Proof Since

$$\mathcal{E} = \frac{1}{2}\,\pi^{-s/2}\,\Gamma\left(\frac{s}{2}\right)\,\zeta(s),$$

we have

$$\log\mathcal{E} = \log\frac{1}{2} - \frac{s}{2}\log\pi + \log\Gamma\left(\frac{s}{2}\right) + \log\zeta(s).$$

Differentiating gives

$$\frac{\mathcal{E}'}{\mathcal{E}} = -\frac{1}{2}\,\log\pi + \frac{1}{2}\,\psi\left(\frac{s}{2}\right) + \frac{\zeta'(s)}{\zeta(s)}.$$

□

Set

$$s = \frac{1}{2} + \sqrt{\lambda}.$$

Then

$$\frac{\Phi'(\lambda)}{\Phi(\lambda)} = \frac{1}{2\sqrt{\lambda}}\,\frac{\mathcal{E}'\left(1/2 + \sqrt{\lambda}\right)}{\mathcal{E}\left(1/2 + \sqrt{\lambda}\right)}.$$

Using the previous lemma,

$$\frac{\Phi'(\lambda)}{\Phi(\lambda)} = \frac{1}{2\sqrt{\lambda}} \left[-\frac{1}{2}\log\pi + \frac{1}{2}\,\psi\left(\frac{1}{4} + \frac{1}{2}\sqrt{\lambda}\right) + \frac{\zeta'\left(1/2 + \sqrt{\lambda}\right)}{\zeta\left(1/2 + \sqrt{\lambda}\right)} \right].$$

For

$$\Re\left(\frac{1}{2} + \sqrt{\lambda}\right) > 1,$$

this may also be written as

$$\frac{\Phi'(\lambda)}{\Phi(\lambda)} = \frac{1}{2\sqrt{\lambda}} \left[-\frac{1}{2}\log\pi + \frac{1}{2}\,\psi\left(\frac{1}{4} + \frac{1}{2}\sqrt{\lambda}\right) - \sum_{n=1}^{\infty} \frac{\Lambda(n)}{n^{1/2+\sqrt{\lambda}}} \right].$$

Lemma 4 (Shifted-square equivalence) *The Riemann hypothesis [1–3] is equivalent to the assertion that every zero of* Φ *lies on* $\mathbb{R}_{\le 0}$.

Proof Suppose the Riemann hypothesis holds. Then every zero of $\mathcal{E}$ has the form

$$s = \frac{1}{2} + i\,\gamma$$

with $\gamma \in \mathbb{R}$. Hence

$$\lambda = \left(s - \frac{1}{2}\right)^2 = (i\gamma)^2 = -\gamma^2 \le 0.$$

Thus every zero of Φ lies on $\mathbb{R}_{\le 0}$. Conversely, suppose every zero of Φ lies on $\mathbb{R}_{\le 0}$. Let ρ be a zero of $\mathcal{E}$, and set

$$\lambda_\rho = \left(\rho - \frac{1}{2}\right)^2.$$

By assumption $\lambda_\rho \le 0$, hence $\lambda_\rho = -\gamma^2$ for some $\gamma \in \mathbb{R}$. Therefore

$$\left(\rho - \frac{1}{2}\right)^2 = -\gamma^2,$$

so

$$\rho = \frac{1}{2} \pm i\,\gamma.$$

Thus

$$\Re(\rho) = \frac{1}{2}.$$

Since the zeros of $\mathcal{E}$ are precisely the nontrivial zeros of $\zeta(s)$, the Riemann hypothesis follows. □

Corollary 3 *Assume the zeta determinant realization and the shifted-square identification. The Riemann hypothesis is equivalent to the absence of tachyonic or complex mass-squared modes in the zeta sector of UCFT [6, 7].*

Proof If $M_\zeta^2 \geq 0$ is self-adjoint, determinant zeros occur only at $\lambda = -m_n^2 \leq 0$, hence the corresponding zeta zeros satisfy $\Re(s) = 1/2$. Conversely, a zero away from the critical line gives a λ-value not lying on $\mathbb{R}_{\leq 0}$, which cannot occur as $-m_n^2$ for a positive self-adjoint mass-squared operator. $\square$

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